

Benchmarking unsupervised anomaly detection in brain MRI

A systematic comparison across training regimes and preprocessing pipelines

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5 UAD METHODS | 10 CONFIGURATIONS EACH | 62 EVALUATED RUNS | 0.883 BEST SLICE AUROC

• IN ONE SENTENCE

The pipeline around a model can matter as much as the model itself.

Contrast, training data and preprocessing each reordered our leaderboard. UAD results should be reported across a matrix of conditions, not one configuration per method.

01 BACKGROUND

Is the gain the model, or the pipeline?

Unsupervised anomaly detection (UAD) learns what healthy brain MRI looks like and flags whatever departs from it, so it needs no pixel-level labels [1].

Diffusion-based methods report large gains, but each study trains on different data, contrasts, field strengths and preprocessing [2]. We asked how much of a reported gain belongs to the model.

02 METHODS

Five methods, one protocol: trained on healthy IXI and MOOD scans in the same ten configurations, tested on BraTS 2020 with the same code. The four without native FSL FAST also ran on StRegA's FAST tissue maps in three configurations, 62 runs in total. Metrics follow [2].

03 STUDY DESIGN

One shared protocol for every method

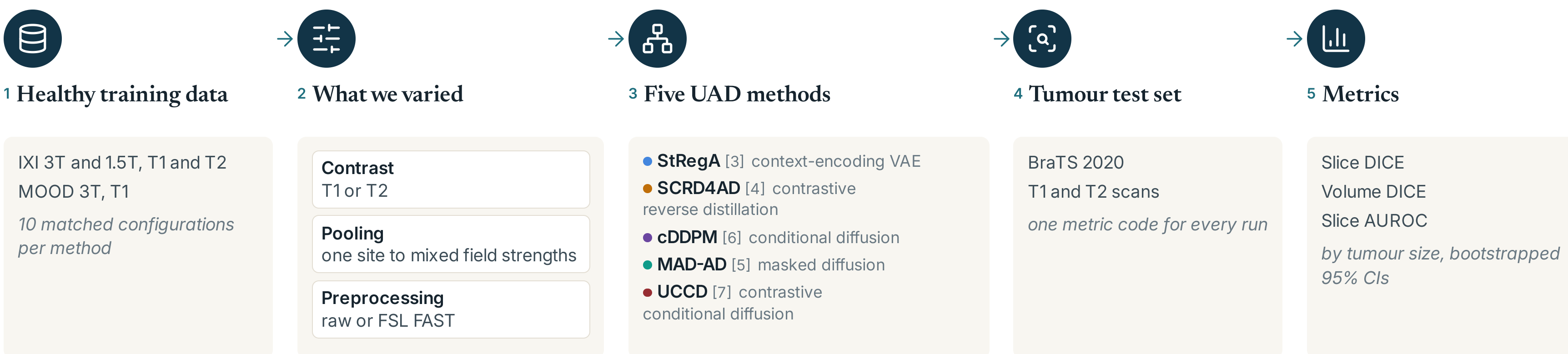


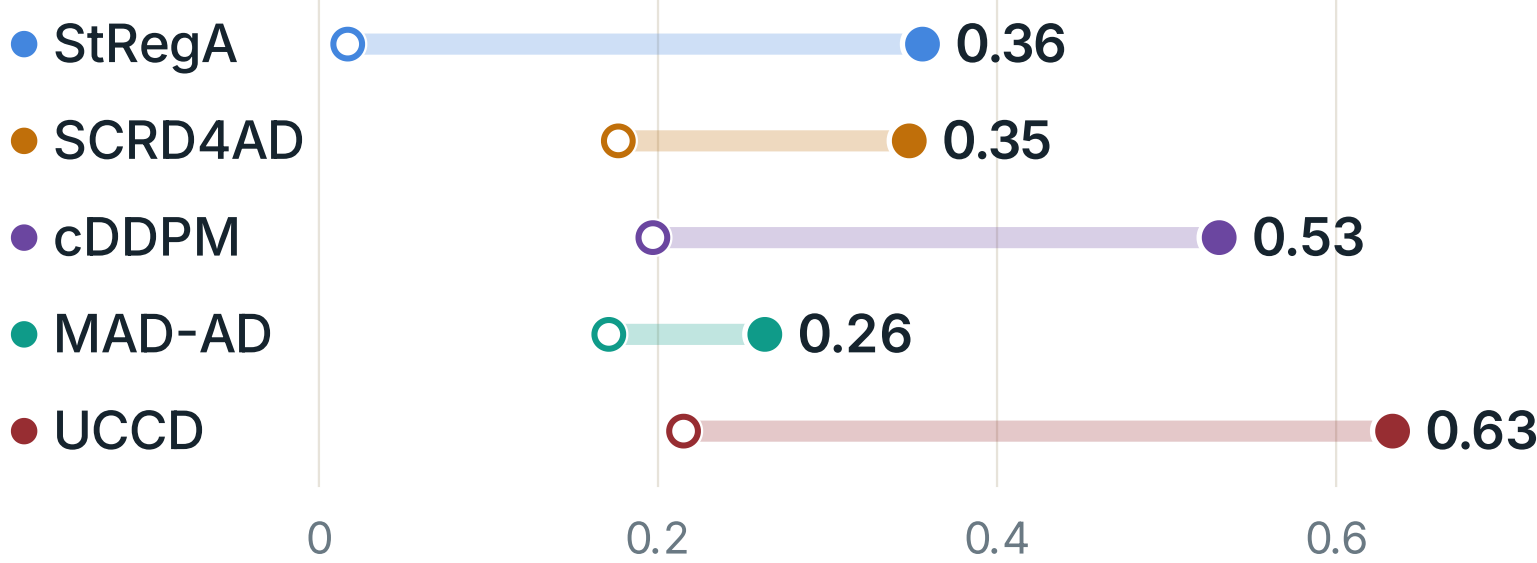
Figure 1. All five methods share the same training configurations, the BraTS 2020 test set and one metric code. The T1 configurations also include MOOD scans; UCCD's T2 runs cover 312 of the 346 test subjects.

F1 CONTRAST

Contrast matters as much as the model

+0.42 best Slice DICE gained by UCCD when trained and tested on T2 instead of T1

Best Slice DICE, trained and tested on
○ T1 ● T2



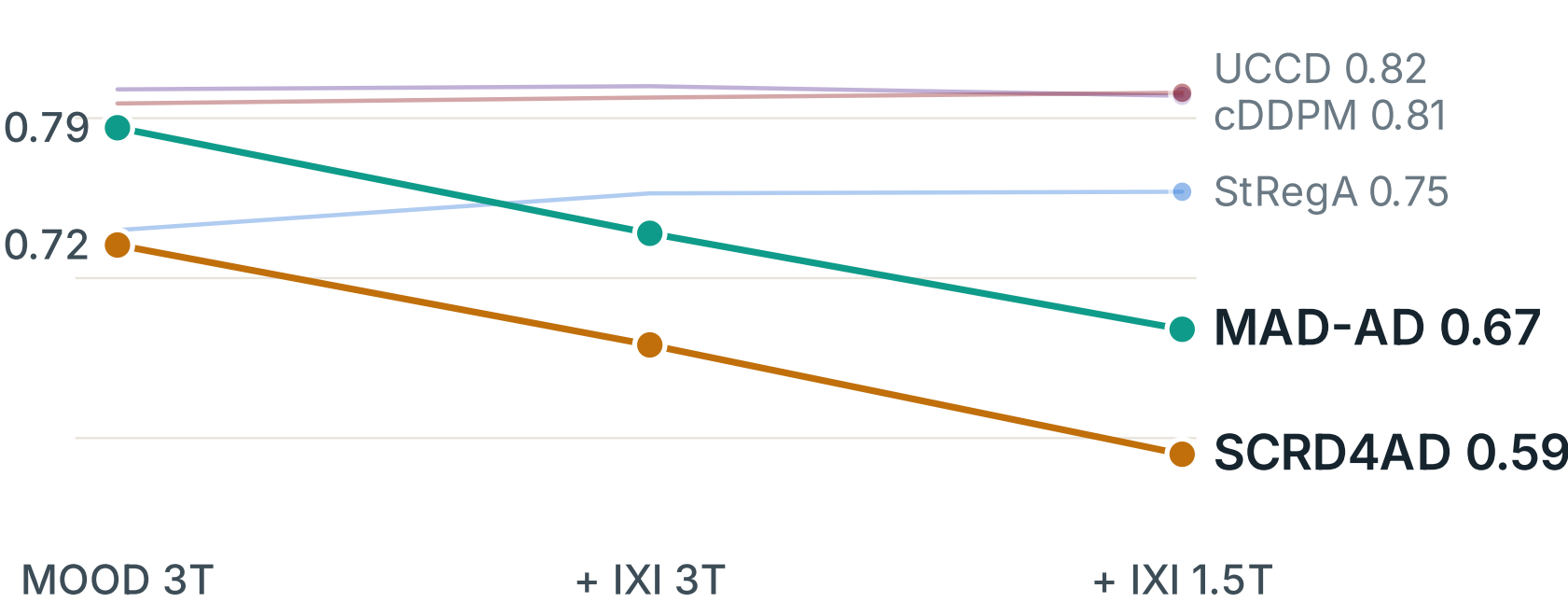
Every method segmented tumours best on T2. Best Slice DICE rose by 0.09 to 0.42 (median 0.33); UCCD's gain exceeds the 0.37 gap between the best and worst method on T2.

F2 POOLING

More training sites are not always better

-0.13 Slice AUROC lost by MAD-AD as IXI 3T, then IXI 1.5T scans joined MOOD

Slice AUROC on BraTS T1 as training sites are added



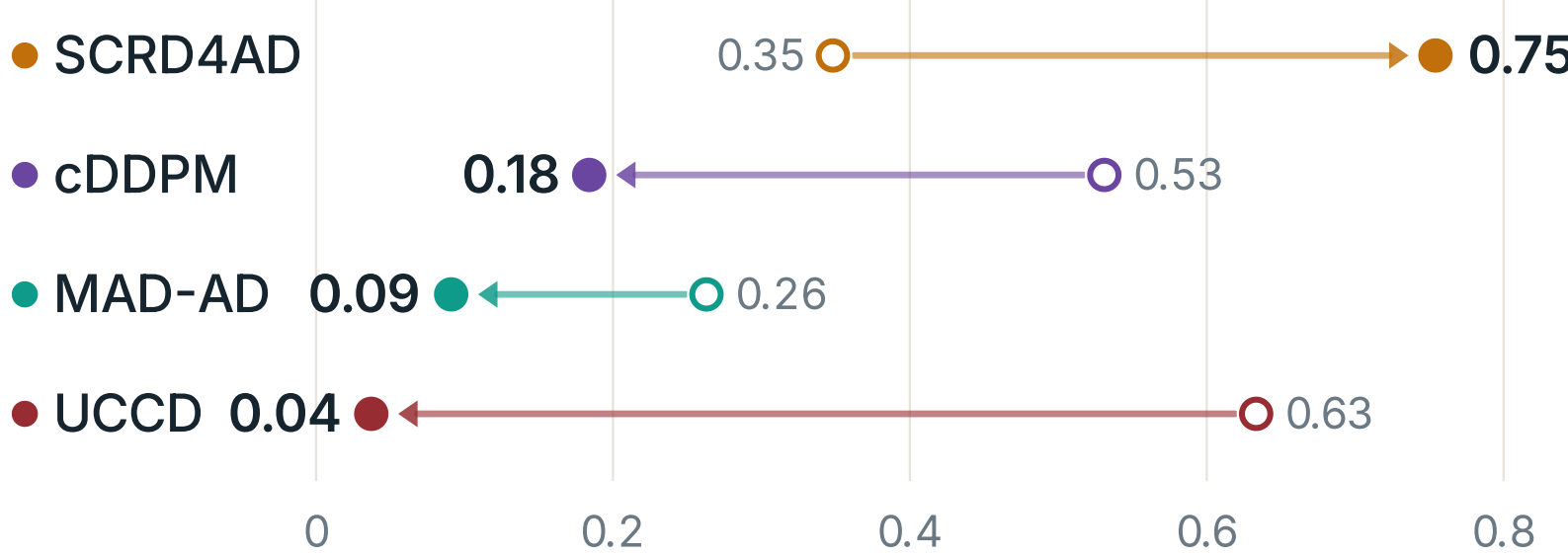
SCR4AD fell by a similar 0.13, while the two conditional diffusion models, cDDPM and UCCD, held steady. How well a method tolerates mixed training data depends on the model.

F3 PREPROCESSING

FAST lifts one model's DICE, sinks three

2.2x SCR4AD Slice DICE on FAST input, against its best raw run

Slice DICE on BraTS T2
○ best raw run ● with FAST input



On StRegA's tissue maps SCR4AD climbed from 0.35 to 0.75. All three diffusion models lost most of their DICE; UCCD, the strongest raw method, fell from 0.63 to 0.04.

04 PREPROCESSING IN DETAIL

Where FAST helps and where it hurts

Volume DICE per BraTS subject, T2 test set

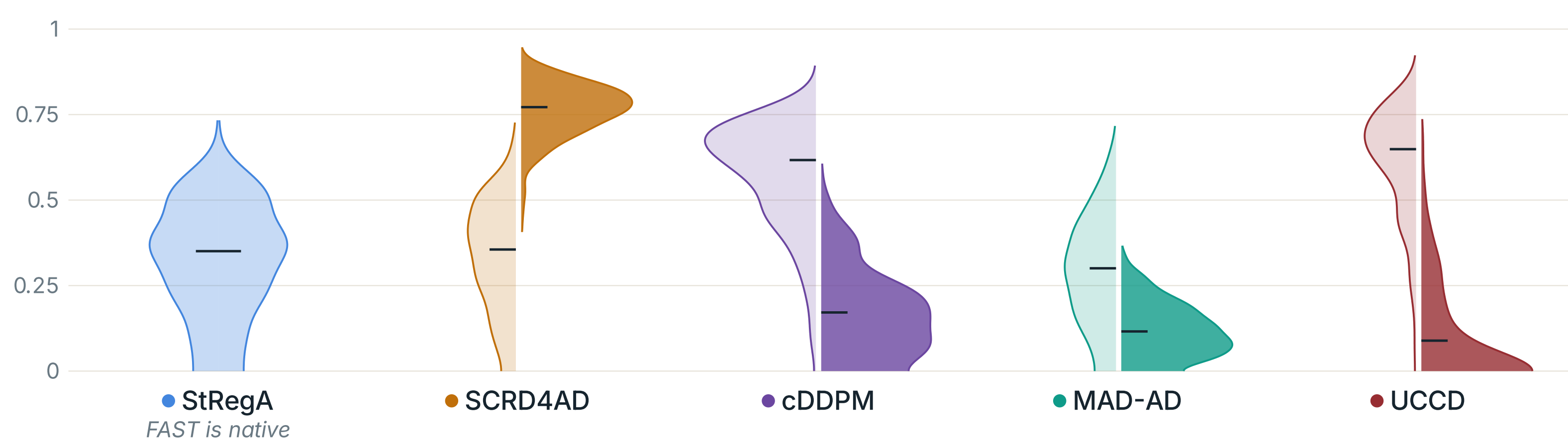


Figure 2. Volume DICE per BraTS 2020 subject (n = 346; 312 for UCCD raw): each method's best raw run (left half) against its FAST run trained on IXI 3T+1.5T (right half), all tested on T2. Lines mark medians. StRegA uses FAST natively.

Our reading

FAST swaps intensities for three tissue classes. That helps SCR4AD outline tumours, but not flag which slices hold one: its Slice AUROC fell from 0.71 to 0.53. The diffusion models lose the intensity detail they reconstruct.

TABLE 1

Best run per method, own pipeline

METHOD · TRAINING → TEST	SLICE DICE	VOLUME DICE	SLICE AUROC	LARGE-TUMOUR DICE
● StRegA IXI 3T+1.5T → T2	0.356	0.351	0.839	0.539
● SCR4AD IXI 3T → T2	0.348	0.356	0.708	0.520
● cDDPM IXI 3T+1.5T → T2	0.531	0.617	0.865	0.768
● MAD-AD IXI 1.5T → T2	0.263	0.301	0.715	0.406
● UCCD IXI 1.5T → T2	0.633	0.649	0.883	0.706

Best run by Slice DICE, before any FAST swap (StRegA's pipeline includes FAST). Bold marks the best value in each column.

Small tumours stay hard

≤0.02

small-lesion DICE in all 60 other runs

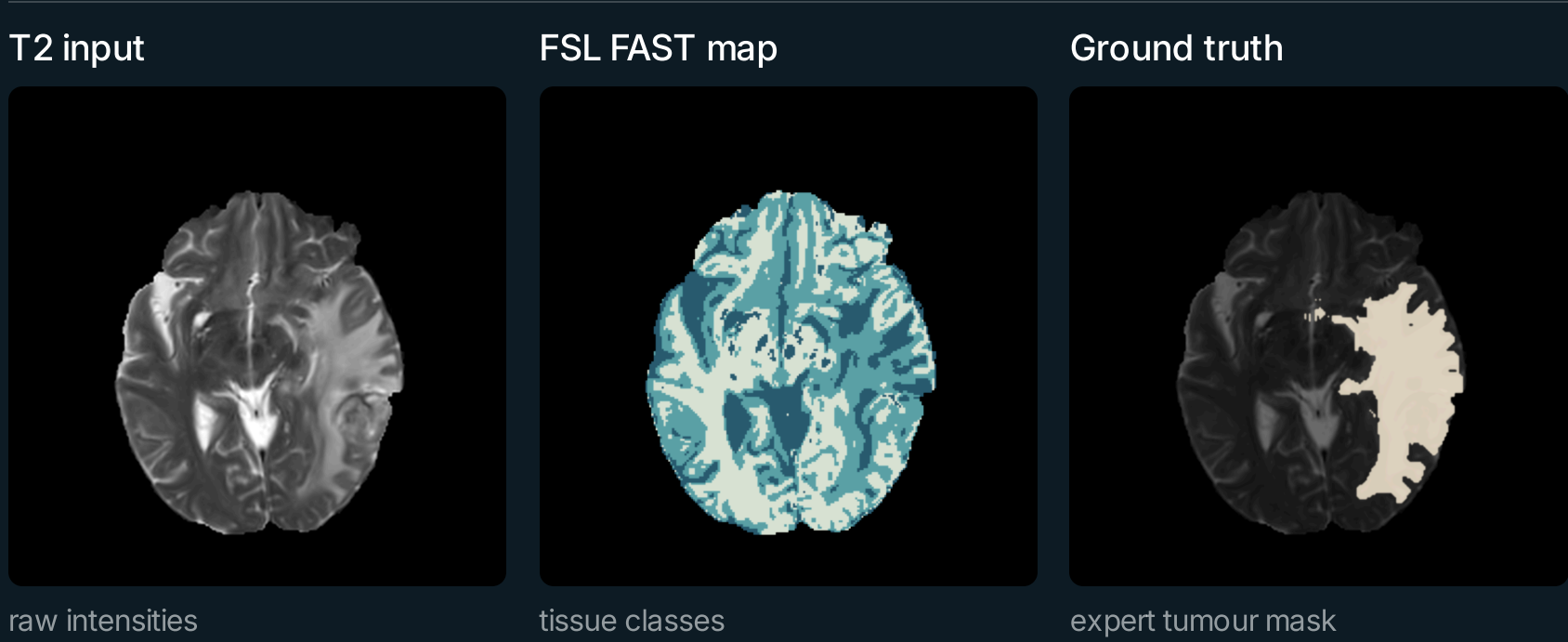
0.26

SCR4AD with FAST on T2 (0.23 on T1), the only exception

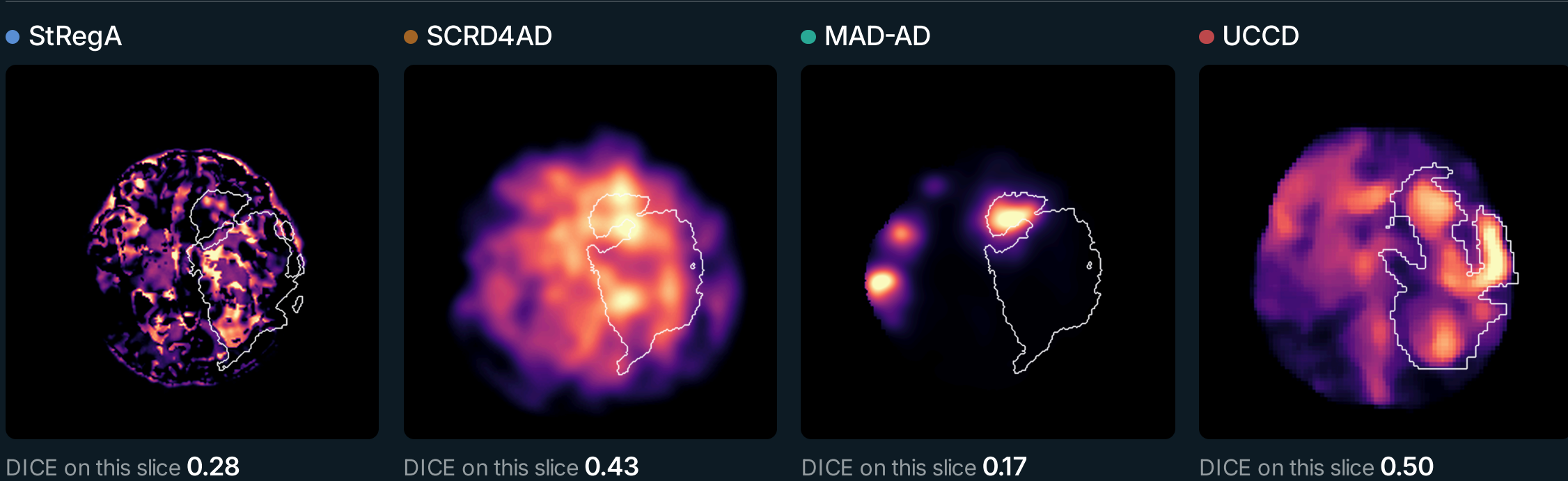
05 ONE CASE

One BraTS case, method by method

INPUTS



BEST RUN, OWN PIPELINE



WITH FAST INPUT

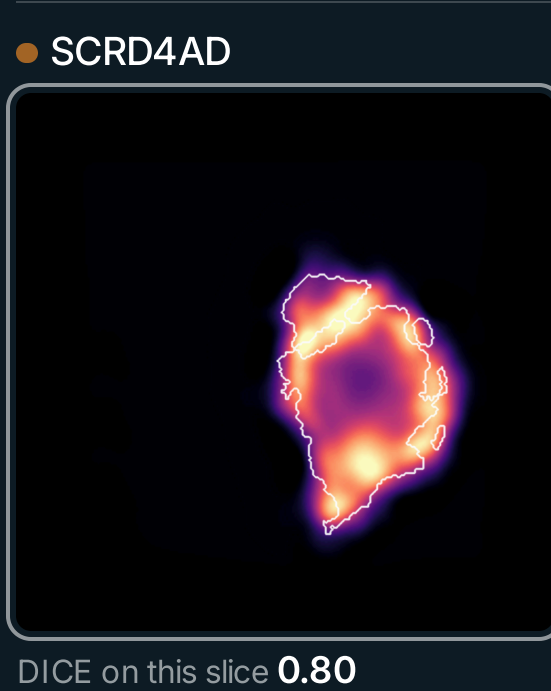


Figure 3. BraTS20. Training. 00149, one axial slice, anterior up. White outline: tumour. Each map comes from its pipeline's own crop and is scaled per panel; numbers are DICE on this slice only. cDDPM exported a non-axial slice for this case, so it is not shown.

LIMITATIONS AND NEXT STEPS

Tested on BraTS 2020 only; further datasets are under way. DICE uses the best threshold on the test labels, as in [1, 6]; StRegA keeps its own Otsu step. All 62 runs and the code are available on request.

REFERENCES

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- [4] Li C et al. *ICLR* 2025. arXiv:2503.13828.

- [5] Beizae F et al. *IPMI* 2025. arXiv:2502.16943.
 - [6] Behrendt F et al. *Comput Biol Med* 2025. arXiv:2312.04215.
 - [7] Patricio C et al. *arXiv*:2406.00772, 2024.
- Data: IXI · MOOD · BraTS 2020, all public.



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